Batch: A2 Roll No.: 16010324044
Experiment / assignment / tutorial No. 8
Grade: AA / AB / BB / BC / CC / CD /DD

Signature of the Staff In-charge with date

TITLE: FILE ALLOCATION STRATEGIES

Objective: Program to implement file allocation stratgies

OUTCOME: Student will be able to Understand the structure and organization of the file system.

A) **SEQUENTIAL:**

AIM:

To write a C program for implementing sequential file allocation method

DESCRIPTION: The most common form of file structure is the sequential file in this type of file, a fixed format is used for records. All records (of the system) have the same length, consisting of the same number of fixed length fields in a particular order because the length and position of each field are known, only the values of fields need to be stored, the field name and length for each field are attributes of the file structure.

ALGORITHM:

Step 1: Start the program.

Step 2: Get the number of files.

Step 3: Get the memory requirement of each file.

Step 4: Allocate the required locations to each in sequential order

a). Randomly select a location from availablelocation s1= random(100);

a) Check whether the required locations are free from the selected location

B) **INDEXED:**

AIM: To implement allocation method using chained method

DESCRIPTION: In the chained method file allocation table contains a field which points to starting block of memory. From it for each bloc a pointer is kept to next successive block. Hence, there is no external fragmentation.

ALGORITHM:

Step 1: Start the program.

Step 2: Get the number of files.

Step 3: Get the memory requirement of each file.

Step 4: Allocate the required locations by selecting a location randomly

q= random(100);

a) Check whether the selected location is free .

b) If the location is free allocate and set flag=1 to the allocated locations.

q=random(100);

{ if(b[q].flag==0) b[q].flag=1; b[q].fno=j;

r[i][j]=q;

Step 5: Print the results file no, length ,Blocks allocated.

Step 6: Stop the program.

C) **LINKED:**

AIM: To implement linked file allocation technique.

DESCRIPTION: In the chained method file allocation table contains a field which points to starting block of memory. From it for each bloc a pointer is kept to next successive block. Hence, there is no external fragmentation

ALGORTHIM:

Step 1: Start the program.

Step 2: Get the number of files.

Step 3: Get the memory requirement of each file.

Step 4: Allocate the required locations by selecting a location randomly q= random(100);

a) Check whether the selected location is free .

b) If the location is free allocate and set flag=1 to the allocated locations. While allocating next location address to attach it to previous location

Signature of faculty in-charge